Count tiny switches in groups

Transistors

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little Wonderer finds 3 transistors, and each one has 3 legs. How many legs are there altogether?

$3 + 3 + 3 =$ _____

Show your thinking.

2. An old machine holds 4 rows of glass tubes, with 5 tubes in each row. How many tubes are there?

$5 + 5 + 5 + 5 =$ _____

Show your thinking.

3. Each time Big Wonderer turns the tap, 4 cups fill up. The big one turns it 2 times. How many cups fill?

$4 + 4 =$ _____

Show your thinking.

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4. A wafer drawing shows 3 rows of tiny chips, with 4 chips in each row. How many chips are there?

$4 + 4 + 4 =$ _____

Show your thinking.

5. The little one lays out 4 rows of switch cards, with 4 cards in each row. How many cards are there?

$4 + 4 + 4 + 4 =$ _____

Show your thinking.

6. They make 5 paper stencils, and each stencil prints 3 shapes. How many shapes are printed?

$3 + 3 + 3 + 3 + 3 =$ _____

Show your thinking.

WONDER DAILY

Transistors

A bead with three legs

A black bead with three thin legs is a transistor. Billions of them work unseen inside the machines around us.

Before transistors

Old machines used hot glass tubes that burned out. In 1947, scientists built the first transistor from gold foil and crystal.

A little turn

Like a tap, a tiny signal on the middle leg opens a path, so a small push controls a much bigger flow.

Made from sand

Silicon is found in sand. A tiny sprinkle of other atoms lets it carry electricity only when told. It is a semiconductor.

On and off

A switch is on or off, like a one or a zero. Eight switches can make 256 different patterns.

Billions on a chip

A chip can hold billions of transistors, printed with light through stencils. Some steer electricity with an unseen push.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Explain a switch clue in your words

Transistors · Curiosity day

Name _____ Date _____

Choose one picture. Say your idea first, then draw or write it.



Before transistors



A little turn



On and off

Draw a transistor with its three legs. Mark the middle leg and draw the path it opens.

transistor · signal · silicon · switch · because · maybe

Transistors replaced glass tubes because … what went wrong with tubes?

A transistor is like a tap because … what does a small turn do?

On and off can make patterns because … what do one and zero mean?

Does your sentence tell what you noticed and use because or maybe to explain?



A Gate With No Hinges

Little Wonderer found a small bug on the floor,
with three skinny legs, and no head, and no more.
"It's no bug," said the big one. "It just looks that way.
It's a transistor, flipping on, off, all the day."

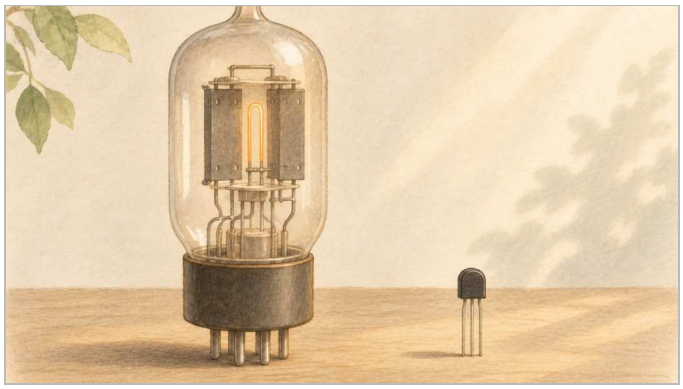
It hasn't a lever. It won't make a click.
It's quiet as pudding, and quick, oh, so quick!
One teeny-small nudge on the leg in the middle
lets a big river through. (It's a bit of a riddle.)

Big Wonderer said, "In one phone, I am told,
there are billions of these, far more than you'd hold.
They don't count to ten, and they don't count by twos.
They count 'on' and 'off' — that's the only way they choose."

So the little one whispered and gave it a pat:
"You're a gate with no hinges. Imagine that!"
Then flipped every lamp in the house on and off,
saying "one," saying "zero," till the big one said, "Off!"

Transistors

01 What came before transistors?



A hot glass tube, and the tiny part that replaced it.

Look closer. Compare the tall glass tube with the tiny black bead beside it.

A little explanation. Glass vacuum tubes. A tube was about as tall as a hand. It glowed, got hot and burned out like an old light bulb. One early computer needed about 18,000 tubes and filled a room. In 1947, three scientists built the first transistor from gold foil pressed onto a small crystal.

Try this. Hold up your hand: an old tube was about that tall. Now look at your smallest fingernail. A transistor you can hold is smaller still. Find one thing in the room the size of each.

02 How can a tiny push control a big flow?



A small turn lets a big flow through.

Look closer. Look at how little the tap handle has turned, and how big the stream is.

A little explanation. A transistor has three legs. Electricity flows in one leg and out another, and the middle leg works like a tap handle. A tiny signal there opens the path for a much bigger flow. The name joins two words: transfer and resistor.

Try this. At a sink, turn the tap handle just a little. Watch how a small turn lets a big stream through. Then turn it back and watch the stream stop.

03 How can on-off switches count?



Light side on, dark side off: a row of switches.

Look closer. Count the light cards and the dark cards in the row.

A little explanation. A transistor switch is on or off, and a computer treats on as one and off as zero. Each extra switch doubles the patterns. Eight switches can make 256 patterns, enough for every letter on a keyboard.

Try this. Lay two paper scraps in a row: light side up is on, dark side up is off. How many different rows can you make? Add a third scrap and count again. Did the number double?

04 How do billions fit on one chip?



Too small to see: billions of tiny switches.

Look closer. Look through the lens at the tiny pattern on the chip.

A little explanation. They are printed, not built one by one. Light shines through a stencil onto a silicon wafer, layer after layer. In many tiny transistors, the middle part never touches the path. It sits behind a thin wall and steers the flow with an unseen electric push.

Try this. Cut a simple shape out of paper to make a stencil. Lay it on another sheet and rub a crayon across it. Print it three times. Chip makers print with light in a similar way.